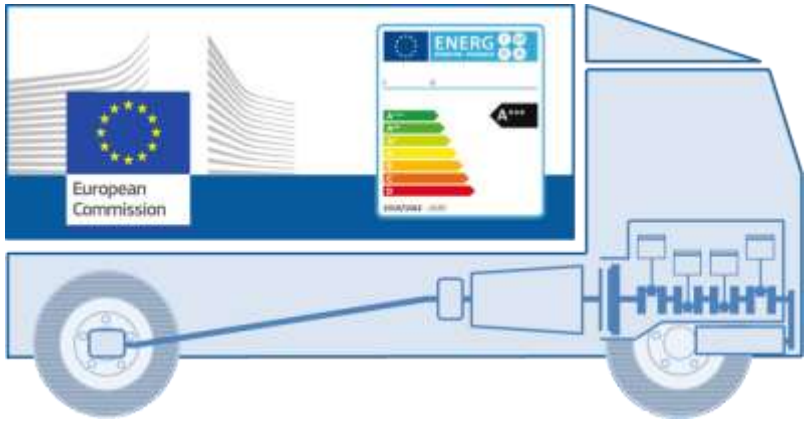
VECTO Engine pre-processing tool

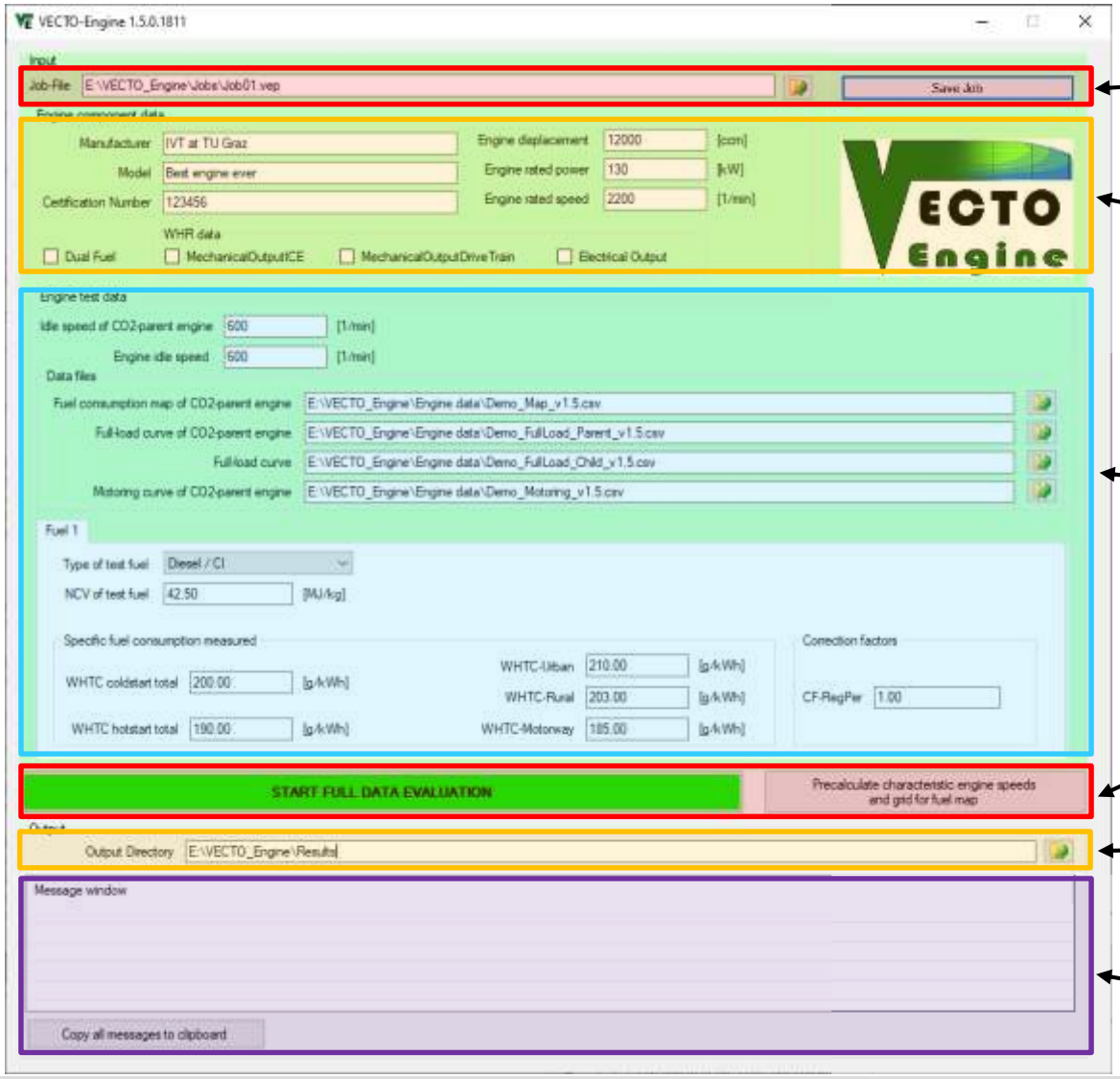
Short documentation and user manual

App Version 1.5.*

01.06.2022



VECTO Engine – GUI overview



Job-file block

Displays path and filename of job file

Open and save job file

Component block

Fields for component information

Checkboxes for technologic features

Test data block

Fields for component data

Path and filename of test data files

Test fuel data

Specific fuel consumption figures

Specific WHR work figures (for engines with WHR systems)

Control button block

Start data evaluation routines

Output block

Displays path of output directory

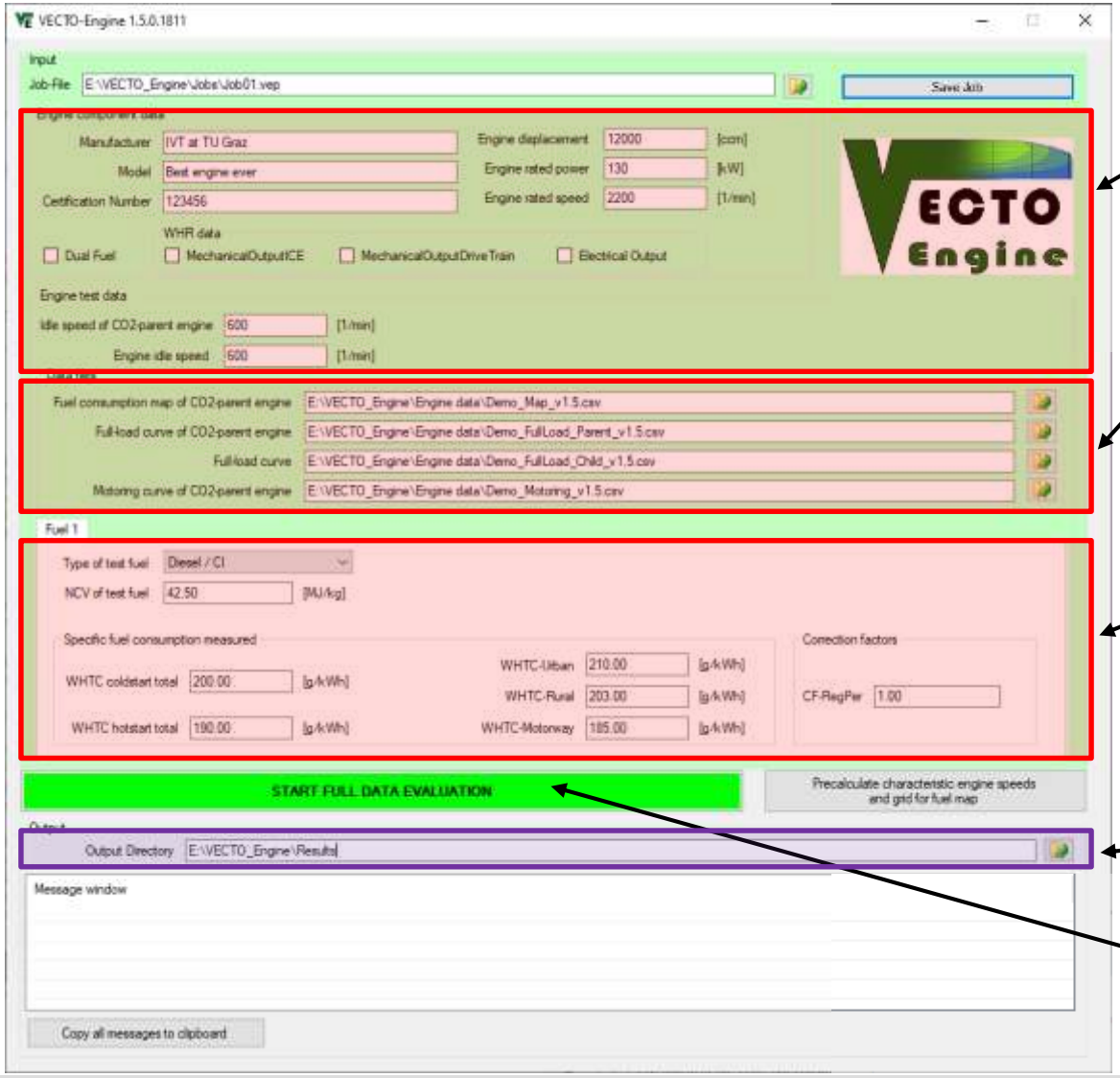
Browse for output folder

Message window block

Displays messages during execution

Export messages to clipboard

VECTO Engine – Operation via GUI



Step 1

Input all component data

Step 2

Load all required input files via separate open file buttons
(for file format please see demo files)

Step 3

Input measured specific FC values and correction factor

→ separate tabs appear depending on setting of technology checkboxes under „engine component data“

OPTIONAL INPUTS:

- *Measured specific WHR work (for engines with WHR systems)*
- Measured specific FC values and correction factor for secondary fuel (for dual-fuel engines)

Step 4

Define directory where output files are stored

Step 5

Press „Start“ button for full evaluation

(OR press other button to precalculate grid for fuel map from CO2-parent full-load curve)

VECTO Engine – Basic features

- Written in VB.net
- Portable application (i.e. no installation required)
- File format conventions *[§6 of Annex V of Regulation (EU) 2017/2400]*
  - CSV (Comma Separated Values) file type
    - List separator is Comma “,”
    - Decimal separator is Dot “.”
- According to Regulation (EU) 2017/2400 the use of VECTO Engine is mandatory
- Output is one single XML file
  - File already hashed
  - Ready for use in VECTO vehicle simulation for declaration

VECTO Engine – Input files

- VECTO Engine needs 4 separate input files
  (determined according to the current version of the technical annex):
  - Fuel consumption map of CO2-parent engine
  - Full-load curve of CO2-parent engine
  - Full-load curve of engine to be certified
  - Motoring curve of CO2-parent engine

- Input file formats
  - For file formats please refer to the technical annex or supplied demo files!
    [§6 of Annex V of Regulation (EU) 2017/2400]

- Demo data supplied
  - Two different full-load curves for two power ratings of one engine CO2-family supplied
    - Child: full-load of lower power rating
    - Parent: full-load of highest power rating
  - Four different fuel consumption map files plus respective job files (*.vep) supplied
    - Standard single fuel engine
    - Dual-fuel engine w/o WHR system
    - Dual-fuel engine w/ WHR system w/ mechanical output to vehicle's drivetrain
    - Dual-fuel engine w/ WHR system w/ electrical output

VECTO Engine – Output files

- VECTO Engine produces one XML output file:
  - File name is created based on GUI input data according to the following scheme:

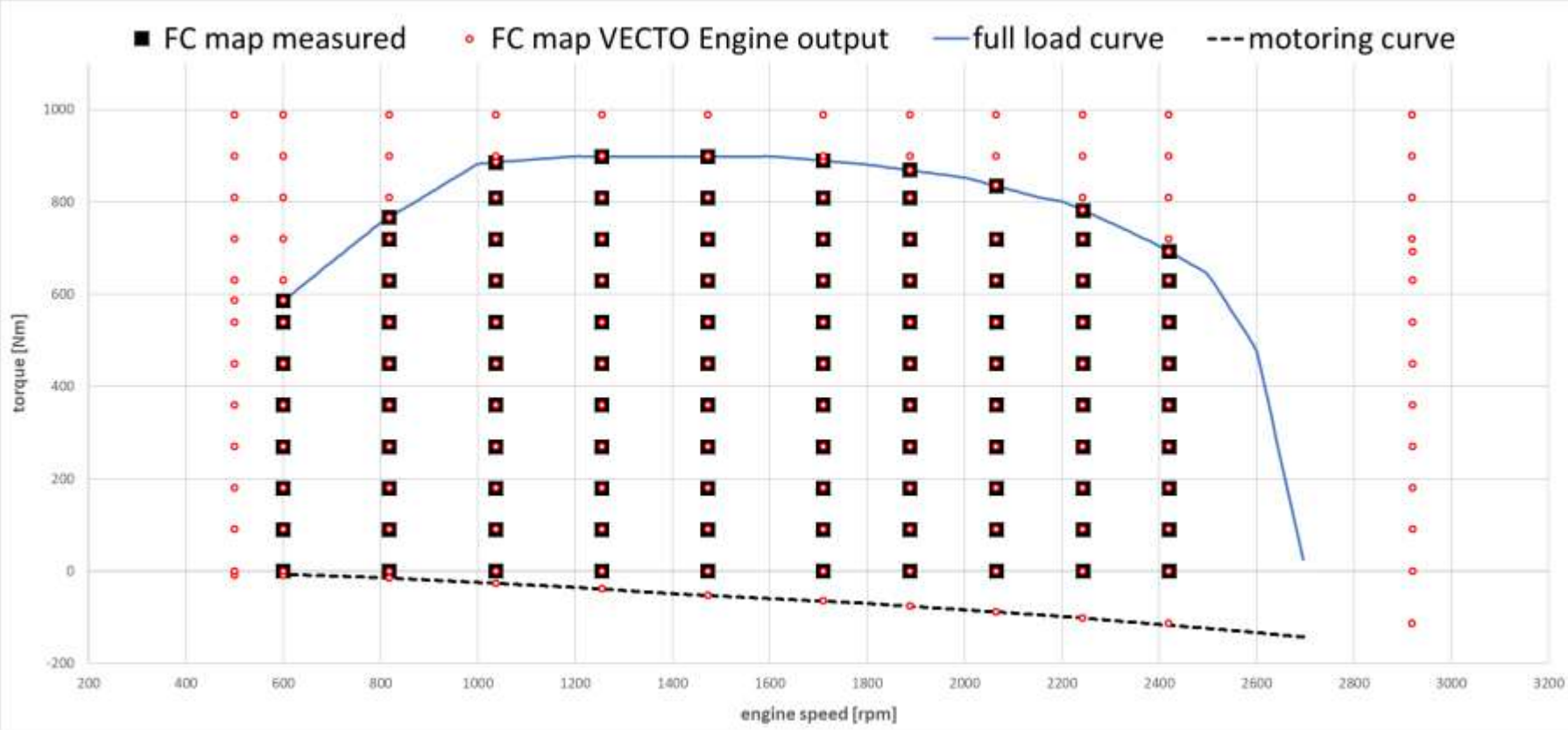
    “Manufacturer” & "_" & “Model” & ".xml“
  - File already hashed
  - Ready for use in VECTO vehicle simulation for declaration

VECTO Engine – Evaluation algorithms performed (p1/2)

For a detailed description of all operations performed during data evaluation please refer to Appendix 8 to Annex V of COMMISSION REGULATION (EU) 2017/2400

VECTO Engine – Evaluation algorithms performed (p2/2)

Example of data processing through VECTO Engine for FC map



VECTO Engine – Further important remarks

- Messages displayed
  - Clear explanations for all relevant errors (ref. to regulation or indication of datapoint)
  - Warnings vs. Errors:
    - A valid XML results file is only created if no errors occur (non-conformity with requirements in regulation)
    - Warnings give a hint that something might be wrong with the input data
      → XML result file is valid despite warnings
      → BUT: warnings should be investigated together with TAA / technical service

- Only one single XML file is created as output
  - File already hashed
  - Ready for use in VECTO vehicle simulation for declaration

Contact data

- All questions, user support and bug reports need to be handled via the JIRA issue tracking in the CITnet service of the EU-COM:
  https://citnet.tech.ec.europa.eu/CITnet/jira/projects/VECTO/issues

- Also future updates of the tool will be released only via this platform:
  https://citnet.tech.ec.europa.eu/CITnet/confluence/display/VECTO/VECTO+Engine+Releases

- If you do not already have an account for this service please contact:
  JRC-VECTO@ec.europa.eu